

$$l(0) = \frac{1}{2} (f_0(X_{data}) - Y_{data})^2$$

$$\begin{aligned} \frac{\partial l}{\partial b_i} &= (f_0(X_{data}) - Y_{data}) \cdot \frac{\partial y_L}{\partial b_i} \\ &= (y_L - Y_{data}) \cdot \frac{\partial y_L}{\partial y_i} \frac{\partial y_i}{\partial b_i} \\ &= \frac{\partial l}{\partial y_L} \cdot \frac{\partial y_i}{\partial b_i} \end{aligned}$$

$\frac{\partial l}{\partial y_L} = \frac{\partial l}{\partial y_{\text{ell}}} = \frac{\partial l}{\partial y_{\text{ell}}} \cdot \frac{\partial y_{\text{ell}}}{\partial b_{\text{ell}}}$   
 $\frac{\partial l}{\partial y_i} = \frac{\partial l}{\partial y_{\text{ell}}} \cdot \frac{\partial y_{\text{ell}}}{\partial y_i}$   
 $\frac{\partial l}{\partial y_i} = \frac{\partial l}{\partial y_{\text{ell}}} \cdot \frac{\partial y_{\text{ell}}}{\partial y_i}$

$$y_i = A_i y_{i-1} + b_i$$

$$\begin{aligned} \frac{\partial l}{\partial A_i} &= \frac{\partial l}{\partial y_i} \cdot \frac{\partial y_i}{\partial A_i} \\ &= \frac{\partial l}{\partial y_i} \cdot y_{i-1} \end{aligned}$$

$$\frac{\partial l}{\partial A_i} = \frac{\partial l}{\partial y_L} \cdot \frac{\partial y_L}{\partial A_i}$$

$$i=L$$

$$= \frac{\partial l}{\partial y_L} \cdot \text{diag}(\sigma'(A_L y_{L-1} + b_L)) \left( \frac{\partial y_L}{\partial y_L} \right)^T y_{L-1}^T$$

$$= \text{diag}(\sigma'(A_L y_{L-1} + b_L)) \left( \frac{\partial l}{\partial y_L} \right)^T y_{L-1}^T$$

$$\frac{\partial l}{\partial y_i} = \frac{\partial l}{\partial y_{i+1}} \frac{\partial y_{i+1}}{\partial y_i}$$